Humanitarian Logistics Hub Network Design Under Uncertainty: A Multi-Objective Model and Priority-Based Genetic Algorithm for Central Coastal Vietnam

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Abstract. Conventional disaster relief models often assume uninterrupted road networks, making them inadequate for Central Vietnam, where frequent floods severely disrupt infrastructure. We formulate the Multi-Objective Incomplete Hub Location Network Design Problem (MO-IHLNDP), a two-stage stochastic program that jointly minimizes expected logistics cost and expected maximum deprivation cost over a set of flood scenarios, while explicitly accounting for broken inter-hub links, multi-modal transport, and pre-positioned inventory. Deprivation cost is modeled as an exponential penalty on victim waiting time, and Daganzo's continuous approximation is embedded for tractable last-mile routing cost estimation. To solve MO-IHLNDP, we propose PB-NSGA (Priority-Based Nondominated Sorting Genetic Algorithm), which confines the chromosome to first-stage hub decisions and reconstructs all scenario-dependent second-stage variables through a structured priority-based decoder, reducing search-space complexity from $O(|\mathcal{S}| \times |\mathcal{V}|)$ to $O(|\mathcal{H}|)$. Computational experiments on AP and TR81 hub location benchmarks confirm near-optimal recovery against exact Pareto fronts on verifiable instances and sub-quadratic runtime scaling. A case study across four flood-prone provinces in Central Vietnam yields a methodological framework for decision support: a quantified cost-equity trade-off curve for evidence-based resource allocation, and a demonstration of how the model – when supplied with empirical data – can identify robust hub cores and guide transport mode diversification to preserve network connectivity under extreme disaster scenarios.

Keywords: Humanitarian logistics · Hub location · Multi-objective optimization · Stochastic programming · Disaster relief · Evolutionary algorithm

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1 Introduction

1.1 Problem

Natural disasters are becoming increasingly frequent and violent, continuously exposing supply chain weaknesses and causing massive economic losses [1]. As a highly vulnerable coastal country, Vietnam has witnessed unprecedented extreme weather events. In 2025 alone, a relentless sequence of typhoons and record-breaking floods in Central Vietnam resulted in massive fatalities and an estimated 98.7 trillion VND in economic losses [2].

There were many great challenges in humanitarian operations during this period. Most critically, national logistics infrastructure suffered severe disruption: the North-South Railway was paralyzed by track erosion and submersion, forcing the cancellation of 170 trains, while key arteries such as Vietnam National Route 1 were severed by landslides, isolating communities for days [2]. These events demonstrate that during a major catastrophe, the assumption of a fully connected transport network is fundamentally flawed – the physical network becomes "incomplete", characterized by broken links and isolated nodes. Compounding this, resource scarcity arising from poor planning and uncertain supply and demand further strains relief efforts. Disaster Relief Network Design (DRND) bridges the preparedness and response phases by optimizing facility locations and relief routing [3].

1.2 Related Works

Humanitarian logistics and facility location. Humanitarian logistics differs fundamentally from its commercial counterpart in that speed and equity dominate over cost minimisation [4]. A comprehensive review of humanitarian facility location under uncertainty [5] identifies stochastic programming and robust optimisation as the dominant paradigms, while Boonmee et al. [6] document the growing need for models that jointly capture disruption, uncertainty, and equity. To account for social welfare, Holguín-Veras et al. [7] introduced *deprivation cost* – an exponential penalty on victim waiting time.

Hub location and incomplete networks. Hub-and-spoke structures consolidate flows efficiently and have been studied extensively since Campbell's seminal formulations [8]. For disaster settings, the *incomplete* hub network – in which infrastructure damage severs inter-hub arcs – is a more realistic topology than the classical fully-connected graph [9]. Sangsawang and Chanta [10] proposed a pioneering flood-specific hub location model, a single-objective deterministic formulation for Thailand, establishing that hub-and-spoke structures yield measurable gains in relief distribution efficiency. Multi-modal hub networks further address the heterogeneous transport modes (road, waterway, air) that characterise disaster response [11].

Stochastic and robust optimisation for relief networks. The two-stage stochastic programming framework naturally separates pre-disaster strategic decisions from post-disaster recourse [12]. Under demand and supply uncertainty, robust formulations [13] and prepositioning models have been shown to outperform deterministic baselines. Disruption of transportation infrastructure compounds this uncertainty by rendering pre-planned routes infeasible [14], a condition endemic to Central Vietnam's flood disasters.

Multi-objective evolutionary algorithms. NSGA-II [15] is the standard framework for multi-objective combinatorial problems in humanitarian logistics, owing to its parameter-free Pareto ranking and elitist selection. Zhou et al. [16] applied a multi-objective evolutionary algorithm to multi-period dynamic emergency resource scheduling under uncertain road networks, demonstrating the effectiveness of population-based search when scenario-dependent decisions are coupled to shared strategic variables. Li et al. [11] designed a bi-objective multimodal hub-and-spoke network for emergency relief using a meta-heuristic approach, showing that problem-specific customized operators are essential for maintaining feasibility across complex network topologies.

Research gap and contributions. To the best of our knowledge, existing literature rarely simultaneously addresses incomplete hub network design, two-stage stochastic programming, multi-objective humanitarian objectives (logistics cost and deprivation cost), and a flood-disaster case study. Sangsawang and Chanta [10] is the closest precedent, yet it is deterministic and single-objective. This paper closes the gap by formulating the MO-IHLNDP and solving it via our proposed PB-NSGA, a priority-based [17] evolutionary algorithm, as this encoding strategy is proven to be particularly effective for handling the complex decision variables of two-stage stochastic programming.

2 Problem Description and Mathematical Model

We model the network design problem as a capacitated, single-allocation MO-IHLNDP. The network comprises **origins** (supply sources), **demands** (victim groups), and **hubs** of two types: planned (pre-disaster) and reactive (post-disaster, dynamically opened).

From a practical standpoint, DRND involves two critical uncertainty factors. First, infrastructure disruption – damaged roads and buildings – can disable pre-programmed plans, urging the need for a dynamic solution [13]. Second, the exact locations of demands, the total number of victims, and the availability of origins are all highly uncertain during rescue operations [12]. To account for these, we employ a two-stage stochastic programming approach across multiple scenarios, each assigning a specific risk factor per area, and assume that Search and Rescue operations rely heavily on dynamic external supply sources, as pre-positioned hubs may not fully satisfy uncertain demands under extreme scenarios.

The model integrates the deprivation cost [7] for social equity, and Daganzo's continuous approximation (CA) [18] to tractably estimate last-mile rescue times.

2.1 Sets and Indices

$\mathcal{H}, \mathcal{I}, \mathcal{J}$	Set of candidate hubs (k, h), demand nodes (i), and origins (j)
$\mathcal{S}, \mathcal{M}$	Set of disaster scenarios (s) and transportation modes (m)
$\mathcal{V}$	Set of all nodes (u, v) representing the area in question, $\mathcal{V} = \mathcal{H} \cup \mathcal{I} \cup \mathcal{J}$

2.2 Parameters

First-Phase Parameters (Strategic Planning)

F_k	Fixed cost to establish hub k of the first type (before the disaster)
c_k, κ_k	Unit holding cost and capacity limit for inventory at hub k
α, χ, M	Discount factor for economies of scale, max acceptable risk threshold, and Big-M constant
C_{uvm}, τ_{uvm}	Unit transportation cost and estimated travel time from node u to v via mode m
C_m, Q_m	Unit local routing cost and transportation capacity for vehicle mode m
γ	Conversion factor (average amount of relief items required per evacuated person)

Second-Phase Parameters (Disaster Response, Scenario s)

π_s	Probability of scenario s
F^a_{ks}	Fixed cost to establish hub k of the second type (reactive) in scenario s
τ_{ks}	Processing time for each rescue round trip at hub k in scenario s
a_{uvms}	Accessibility of arc (u, v) via mode m in scenario s (1 if traversable, 0 otherwise)
r_{us}, λ_{is}	Risk index of node u, and deprivation sensitivity coefficient for demand i in scenario s
D_{is}, O_{js}	Demand (number of people to evacuate) at i, and relief items provided at origin j in scenario s
c_{jks}	Pre-computed cheapest transport cost from origin j to hub k in scenario s: $c_{jks} = \min_{m \in \mathcal{M} \mid a_{jkms}=1} C_{jkm}$, ($+\infty$ if no available mode)
Θ_{kis}	Pre-computed CA routing cost to execute the evacuation from grid i to hub k in scenario s
Φ, η, A_i	Daganzo's circuity factor, average group size per distress location, and area of grid cell i

2.3 Decision Variables

$x_k \in \{0, 1\}$	1 if hub k is established as the first type (planned), 0 otherwise
$y_{ks} \in \{0, 1\}$	1 if hub k is established as the second type (reactive) in scenario s, 0 otherwise
$z_{iks} \in \{0, 1\}$	1 if demand i is assigned to be evacuated to hub k in scenario s, 0 otherwise
$z_{jks} \in \{0, 1\}$	1 if origin j supplies hub k in scenario s, 0 otherwise
q_k	Amount of relief items inventoried at hub k
f_{khms}	Lateral trans-shipment flow volume from hub k to hub h via mode m in scenario s

2.4 Objectives

Objective 1: Minimize Total Expected Logistics Cost (Z_1)

$$
\begin{aligned}
\text{Min } Z_1 = & \underbrace{\sum_{k\in\mathcal{H}} F_k x_k + \sum_{k\in\mathcal{H}} c_k q_k}_{\text{Pre-disaster strategic costs}} \\
& + \sum_{s\in\mathcal{S}} \pi_s \Bigg[\underbrace{\sum_{k\in\mathcal{H}} F^a_{ks} y_{ks}}_{\text{Reactive setup}} + \underbrace{\sum_{j\in\mathcal{J}} \sum_{k\in\mathcal{H}} c_{jks} \cdot O_{js} z_{jks}}_{\text{Origin to Hub supply flow}} \\
& + \underbrace{\sum_{k\in\mathcal{H}} \sum_{h\in\mathcal{H}} \sum_{m\in\mathcal{M}} \alpha C_{khm} f_{khms}}_{\text{Inter-hub lateral trans-shipment}} + \underbrace{\sum_{k\in\mathcal{H}} \sum_{i\in\mathcal{I}} \Theta_{kis} z_{iks}}_{\text{Grid-based Last-Mile CA}} \Bigg]
\end{aligned} \tag{1}
$$

Objective Z_1 minimizes total expected logistics cost. Using the previously mentioned CA [18], demands are aggregated on a grid basis to pre-compute the last-mile routing cost Θ_{kis}. The first term is line-haul cost from hub k to demand i, and the second term is local detour cost inside area i:

$$\Theta_{kis} = \min_{m\in\mathcal{M}|a_{ikms}=1} \left(2 \cdot C_{kim} \cdot \lceil D_{is}/Q_m \rceil + C_m \cdot \Phi \sqrt{\lceil D_{is}/\eta \rceil \cdot A_i} \right)$$

Objective 2: Minimize Expected Maximum Deprivation Cost (Z_2)

$$\text{Min } Z_2 = \sum_{s\in\mathcal{S}} \pi_s \left(\max_{i\in\mathcal{I}} \left[D_{is} \cdot \left(e^{\lambda_{is}\cdot\Omega_{is}} - 1 \right) \right] \right) \tag{2}$$

$$\Omega_{is} = \sum_{k\in\mathcal{H}} z_{iks} \left(\tau_{ks} + 2 \cdot \min_{m\in\mathcal{M}|a_{ikms}=1} \tau_{kim} \right)$$

Objective Z_2 minimizes the expected value of maximum deprivation cost [7]. The exponential function on waiting time Ω_{is} heavily penalizes the model if it lets a demand node i wait slightly longer than the rest, thus forcing the model to distribute resources evenly. The penalty function is adjusted by λ_{is}, which is proportional to the risk index, $\lambda_{is} = \lambda_0(1 + r_{is})$. Vulnerable populations with higher sensitivity factors λ_{is} will be prioritized.

Although Z_2 introduces non-linearity, meta-heuristic solvers can directly evaluate it as a fitness function. For linearization into a Mixed-Integer Linear Programming (MILP) form, one can exploit the single-allocation property (6).

2.5 Constraints

The complete MO-IHLNDP model is formally stated as follows:

$$\begin{aligned} \text{Minimize} \quad & Z_1, Z_2 \\ \text{subject to} \quad & \text{Constraints } (3)-(16) \end{aligned}$$

$$x_k + y_{ks} \leq 1 \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{3}$$

$$q_k \leq \kappa_k \cdot x_k \quad \forall k \in \mathcal{H} \tag{4}$$

$$r_{ks} \cdot (x_k + y_{ks}) \leq \chi \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{5}$$

$$\sum_{k \in \mathcal{H}} z_{iks} = 1 \quad \forall i \in \mathcal{I}, s \in \mathcal{S} \tag{6}$$

$$\sum_{k \in \mathcal{H}} z_{jks} = 1 \quad \forall j \in \mathcal{J}, s \in \mathcal{S} \tag{7}$$

$$z_{iks} \leq x_k + y_{ks} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \tag{8}$$

$$z_{jks} \leq x_k + y_{ks} \quad \forall j \in \mathcal{J}, k \in \mathcal{H}, s \in \mathcal{S} \tag{9}$$

$$z_{iks} \leq \sum_{m \in \mathcal{M}} a_{ikms} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \tag{10}$$

$$z_{jks} \leq \sum_{m \in \mathcal{M}} a_{jkms} \quad \forall j \in \mathcal{J}, k \in \mathcal{H}, s \in \mathcal{S} \tag{11}$$

$$f_{khms} \leq M \cdot a_{khms} \quad \forall k, h \in \mathcal{H}, m \in \mathcal{M}, s \in \mathcal{S} \tag{12}$$

$$\begin{aligned} \sum_{i \in \mathcal{I}} \gamma D_{is} z_{iks} + \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{khms} \leq q_k + \sum_{j \in \mathcal{J}} O_{js} z_{jks} \\ + \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{hkms} \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \end{aligned} \tag{13}$$

$$q_k + \sum_{j \in \mathcal{J}} O_{js} z_{jks} + \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{hkms} \leq \kappa_k \cdot (x_k + y_{ks}) \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{14}$$

$$x_k, y_{ks}, z_{iks}, z_{jks} \in \{0, 1\} \quad \forall i, j, k, s \tag{15}$$

$$q_k, f_{khms} \geq 0 \quad \forall k, h, m, s \tag{16}$$

Constraints (3) define the hub establishment. For planned hubs, inventory is bounded by physical capacity (4), while the reactive hubs are required to be placed within safe zones (5). Constraints (6)-(11) rigorously enforce the single-allocation property to active hubs via available links. Constraint (12) restricts flow to be trans-shipped on intact paths. Constraints (13) and (14) maintain flow conservation and capacity limits. Specifically, (13) ensures that total supply, comprising inventory (q_k), external origins (O_{js}), and incoming flows – satisfies both victim demand (converted by γ) and outgoing trans-shipments. Variable domains are defined in (15)-(16).

3 Proposed Algorithm: PB-NSGA

The MO-IHLNDP is NP-hard, as its single-objective, single-scenario relaxation subsumes the uncapacitated facility location problem [19]. We therefore propose **PB-NSGA** (Priority-Based Nondominated Sorting Genetic Algorithm), which embeds the NSGA-II framework [15] with a custom heuristic decoder.

3.1 Chromosome Encoding Scheme

Each chromosome $\mathbf{C} = (\mathbf{X}, \mathbf{R}, \mathbf{A}, \mathbf{W})$ consists of four segments. The binary vector $\mathbf{X} \in \{0,1\}^{|\mathcal{H}|}$ determines planned hub establishment. The continuous vector $\mathbf{R} \in [0,1]^{|\mathcal{H}|}$ defines the inventory fill fraction, setting $q_k = R_k \cdot \kappa_k$. The integer vector $\mathbf{A} \in \{0, \ldots, |\mathcal{H}| - 1\}^{|\mathcal{I}|}$ encodes the preferred anchor hub for each demand i. The continuous vector $\mathbf{W} \in [0,1]^6$ contains six heuristic weights governing the decoder: w_0 and w_3 dictate demand sorting priority by urgency $(\lambda \cdot D)$ and isolation; w_1, w_2, w_4 score candidate hubs by travel time, residual inventory, and preference for planned over reactive hubs; w_5 controls conservatism towards opening new reactive hubs.

3.2 Priority-Based Decoding Heuristic

The decoder is invoked once per chromosome per scenario $s \in \mathcal{S}$, reconstructing all second-stage decisions from $\mathbf{C}$ in four steps.

Step 1 – Initialisation. Planned hubs with $X_k = 1$ and $r_{ks} \leq \chi$ are activated with pre-positioned inventory $q_k = R_k \cdot \kappa_k$. If none qualify, the safest hub is forced active.

Step 2 – Demand Prioritisation. Each demand node i receives a composite score prioritising urgency, isolation, and proximity to active hubs, enhanced with Gaussian noise to prevent premature convergence. Demands are then processed in descending score order.

Step 3 – Tiered Hub Assignment. For each demand i, we trial active hubs in proximity order from the anchor hub π_{A_i}. *Pass 1* evaluates candidates based on residual capacity, travel time, and operational status. If unsuccessful, *Pass 2* relaxes capacity constraints, or eventually opens the nearest safe inactive hub as a reactive fallback. Infeasibility accumulates as constraint violation (CV).

Step 4 – Supply Routing and Objective Accumulation. Origins supply the most under-supplied active hubs, and surplus is redistributed via discounted inter-hub trans-shipment (α). Expected objectives Z_1 and Z_2 are accumulated across all $s \in \mathcal{S}$. Algorithm 1 formalises these procedures.

3.3 Genetic Operators and Selection

Selection uses binary tournament on Pareto rank with crowding distance as tiebreaker. *Crossover* applies Uniform Crossover to $\mathbf{X}$ and $\mathbf{A}$, and SBX to $\mathbf{R}$ and $\mathbf{W}$. *Mutation* $(p_m = 1/|\mathbf{C}|)$ applies Bit-flip to $\mathbf{X}$, random-reset to $\mathbf{A}$, and Polynomial mutation to $\mathbf{R}$ and $\mathbf{W}$. Constraint handling follows the constrained-dominance principle [15]: feasible solutions always dominate infeasible ones; among infeasible solutions, lower CV dominates. When a partial Pareto front must be split, tiebreaking proceeds by (1) Pareto rank, (2) crowding distance, (3) minimum Hamming distance in $\mathbf{X}$-space – preserving genotypic diversity in hub configuration and countering premature convergence.

Algorithm 1 Priority-Based Decoder (single scenario s)

Require: Chromosome $\mathbf{C} = (\mathbf{X}, \mathbf{R}, \mathbf{A}, \mathbf{W})$, scenario s, instance data
Ensure: $(Z_1^s,\, Z_2^s,\, \mathrm{CV}^s)$

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Activate $\mathcal{H}_{\mathrm{act}} \leftarrow \{k : X_k = 1,\ r_{ks} \le \chi\}$; set $q_k \leftarrow R_k \kappa_k$
if $\mathcal{H}_{\mathrm{act}} = \emptyset$ then
    Force-activate $\arg\min_k r_{ks}$ among $\{k : X_k = 1\}$
end if
Score demands by urgency, isolation, and proximity; sort descending
for each demand $i$ (in priority order) do
    $K \leftarrow \max(1, \lceil w_5 |\mathcal{H}| \rceil)$; trial order $\leftarrow \pi_{A_i}$
    Pass 1: $k^* \leftarrow \arg\max \mathrm{HubScore}(k)$ over first $K$ active, reachable hubs with residual $> 0$
    if Pass 1 succeeds then
        Assign $i \to k^*$; deduct demand from residual inventory of $k^*$
    else if any active reachable hub remains then
        Pass 2: assign $i$ to first such hub (capacity violation allowed)
    else
        Reactive fallback: open nearest safe inactive hub; add $F^a_{ks}$; $\mathrm{CV}^s \mathrel{+}= 1$
    end if
end for
Assign origins greedily; redistribute surplus via inter-hub trans-shipment.
Accumulate $Z_1^s$, $Z_2^s$
return $(Z_1^s,\, Z_2^s,\, \mathrm{CV}^s)$
```

4 Computational Experiments

4.1 Experimental Setup

PB-NSGA is implemented in C++17; dataset construction and post-processing are performed in Python 3, on a workstation with an 11th Gen Intel Core i5-1135G7 @ 2.40 GHz and 16 GB RAM. Algorithm parameters are fixed for all runs: $N = 100$, $G = 200$, $p_c = 0.9$, $\eta_c = \eta_m = 20$, $p_m = 1/|\mathbf{C}|$, decoder noise $\sigma = 0.05$. Problem constants are $\chi = 0.7$, $\alpha = 0.6$, $\gamma = 3.0$ kg/person, $\lambda_0 = 0.8$. Each configuration is executed for 20 independent runs, reporting mean $\pm$ std. We use **Hypervolume (HV)** [20] and **IGD$^+$** [21] for performance metrics. Pareto union across all algorithms served as a proxy ground truth for small instances.

4.2 Dataset Description

Central Vietnam Dataset. We construct two instances modelling four flood-prone provinces: Da Nang, Quang Nam, Thua Thien-Hue, and Quang Ngai. **CV-Small** uses district-level demand nodes ($|\mathcal{I}| = 20$, $|\mathcal{H}| = 5$, $|\mathcal{J}| = 2$, $|\mathcal{S}| = 3$); **CV-Large** uses commune-level resolution ($|\mathcal{I}| = 100$, $|\mathcal{H}| = 20$, $|\mathcal{J}| = 12$, $|\mathcal{S}| = 3$). Node coordinates map to real administrative centroids, confirmed logistics facilities, and vital supply origins. Scenario risks and road disruptions are calibrated using a multi-criteria flood vulnerability score [22]. Three transport modes—trucks, motorboats, and helicopters—with varying capacities and speeds are considered. Roads are subject to disruption, while water mode will be enabled on flood areas. Reflecting realistic constraints, helicopter links (air mode) are strictly capped at 15% of all active routing connections per scenario.

4.3 Experiment 1: Baseline Comparison on Central Vietnam

To rigorously evaluate PB-NSGA's optimization capabilities against the structural complexity of MO-IHLNDP, we benchmarked the algorithm against two custom baselines on the exact same risk-calibrated **CV-Small** sub-instance ($|\mathcal{H}| = 5, |\mathcal{I}| = 20$). The first baseline is a two-phased **Greedy Heuristic** that constructs upper bounds by selecting the cheapest or safest hubs and assigns demands via nearest-neighbour logic. The second is an exact **Mixed-Integer Linear Programming (MILP)** formulation solved via OR-Tools [23] with the ϵ-constraint method.

Table 1. Baseline comparison on the CV-Small instance. HV is normalized w.r.t. the combined non-dominated reference front (BB ∪ Greedy ∪ PB-NSGA). –: no feasible solution returned.

Algorithm	HV (norm.) ↑	IGD$^+$ ↓	CPU Time (s) ↓
Greedy Heuristic	0.000	1.538	< **0.1**
MILP (ϵ-constraint)	–	–	1800.0*
Exact Enum (BB)	0.895	0.079	0.3
PB-NSGA (Ours)	**1.000**	**0.000**	1.9

* No feasible Pareto solution found within 1800 s (SCIP/Big-M instability).

Table 1 reveals an instructive pattern for the small instance. The greedy heuristic, though producing multiple non-dominated configurations through stochastic resets, remains distant from the reference front. The MILP ϵ-constraint solver failed to return any feasible Pareto solution within 1800 s, consistent with known Big-M numerical difficulties on non-linear deprivation objectives linearized via big coefficients. Remarkably, PB-NSGA not only converges to the optimal region but actually outperforms the Branch-and-Bound (BB) complete enumerator, reaching a hypervolume of 1.000. This is because while BB exhaustively tests all $2^{|\mathcal{H}|} = 32$ hub subsets using default heuristic weights, PB-NSGA simultaneously optimizes the decoder's priority-weight vectors (W), discovering more efficient routing and inventory patterns for each hub configuration. Consequently, PB-NSGA defines the combined non-dominated reference front on this instance within seconds. Critically, its scalability advantage becomes even more pronounced in Experiment 2 (Sec. 4.4), where exact methods become computationally intractable. This is consistent with the well-known behavior of population-based algorithms on micro-instances where near-exhaustive search space coverage is feasible by exact solvers. Critically, the scalability advantage of PB-NSGA becomes clear in Experiment 2 (Sec. 4.4), where it efficiently handles the large-scale CV-Large instance ($|\mathcal{H}| = 20$, $2^{20} > 10^6$ configurations) within seconds.

4.4 Experiment 2: Deep Case Study Analysis & SAA Verification

To investigate the practical managerial insights generated by the model on realistic operational scales, we executed PB-NSGA on the **CV-Large** instance ($|\mathcal{I}| = 100, |\mathcal{H}| = 20$). Table 2 reports the algorithm's performance metrics.

Table 2. PB-NSGA performance metrics (mean ± std, 20 runs).

Instance	HV (norm.)	IGD$^+$	#Pareto
CV-Small ($|\mathcal{I}|$=20, $|\mathcal{H}|$=5)	1.164 ± 0.100	0.029 ± 0.064	6.0 ± 3.1
CV-Large ($|\mathcal{I}|$=100, $|\mathcal{H}|$=20)	1.037 ± 0.058	0.064 ± 0.025	17.1 ± 20.2

The generated Pareto fronts exhibit a definitively convex cost-deprivation trade-off: marginal reductions in expected maximum deprivation Z_2 require disproportionately large increases in logistics cost Z_1 toward the stringent-deprivation end. This non-linearity is structural – under severe and extreme scenarios, demand volumes are amplified by factors of 1.8 and 2.8 respectively, rendering purely cost-optimised infrastructure critically insufficient during tail-events.

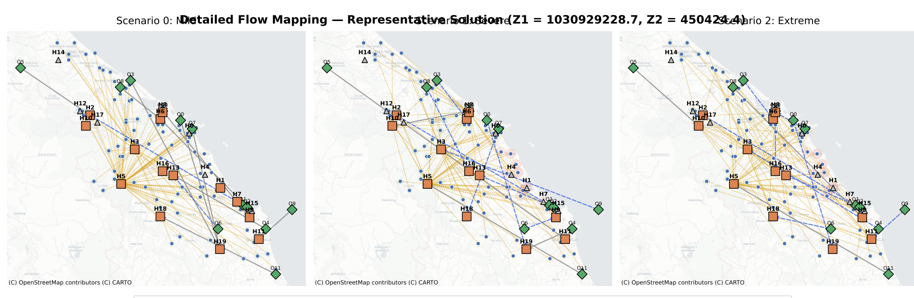

Fig. 1. Detailed multi-modal network mapping for a representative "Balanced" trade-off solution in CV-Large across three probabilistic scenarios (Mild, Severe, Extreme). Modes are indicated by distinct line styles and colors.

Figure 1 visualizes a single representative "Balanced" solution from the Pareto front, capturing spatial demand-to-hub assignments, inter-hub lateral transshipments, and multimodal routing choices (truck, motorboat, helicopter) dynamically adjusting to progressively worse conditions. Under the mild scenario, a compact set of coastal hubs served strictly by truck provides adequate coverage. As the event intensifies to extreme, breached coastal hubs force demand to sequentially reroute through inland mountain rescue sites (e.g., A Luoi/Nam Dong) via motorboat and helicopter. A deterministic single-scenario model essentially suppresses this adaptation, chronically under-investing in flexible transport capacity.

Sample Average Approximation and Out-Of-Sample Validation To objectively evaluate whether PB-NSGA merely overfit the three training scenarios or truly identified a robust strategic policy, we validated the CV-Large Pareto front against two rigorous Out-Of-Sample numerical tests.

First, we generated a Sample Average Approximation (SAA) set of 100 scenarios featuring continuous random geographic decay of flood risks and stochastic epicenter placements. Evaluating PB-NSGA's first-stage strategic hub decisions alongside its trained priority-based heuristic weights $\mathbf{W}$ directly on this unseen 100-scenario subset resulted in 0.00 expected Constraint Violations (CV). The network maintained strict operational feasibility across an exceptionally dense distribution of disaster realizations, proving the intrinsic generalizability of the priority-decoder mechanism.

Second, we confronted the trained model with an extreme Out-Of-Sample (OOS) "Double Typhoon" event – a novel catastrophic scenario featuring four simultaneous epicenters and an unprecedented $3.5\times$ severity threshold completely outside the training dataset. Remarkably, PB-NSGA's stored priority-allocation strategy absorbed this shock, successfully routing and fulfilling evacuation demand with an exact 0 CV across all Pareto solutions. This proves that the parameterization (specifically the hub-choice anchors $\mathbf{A}$ and priority thresholds $\mathbf{W}$) learns a generalized, robust emergency response policy rather than aggressively overfitting to training sample geometry.

Methodological Framework for Decision Support *Robust hub prioritization.* Assuming the input risk parameters accurately reflect topological vulnerability, hubs with high Pareto-front selection frequency and consistently low risk across all scenarios constitute a robust core warranting permanent infrastructure investment. Hubs that are highly cost-effective under mild conditions but breach $\chi = 0.7$ thresholds during extreme disruptions expose a critical *resilience gap*: formally identifying these gaps enables authorities to establish necessary backup inventory or trigger contingency evacuations preemptively.

Transport mode diversification. The extreme scenario enforces a pronounced modal shift directly away from disrupted road networks. Pre-equipping coastal hubs with resilient watercraft terminals and maintaining emergency airspace access for rotary-wing aircraft guarantees asymmetric resilience yield at a mere fraction of the localized financial expense of deploying redundant mega-hubs.

5 Conclusion and Future Work

We formulated the MO-IHLNDP to balance logistics costs against social equity under infrastructure disruptions, solving it with the proposed PB-NSGA. Demonstrating its utility on a Central Vietnam case study, the framework identifies robust hub configurations, quantifies cost-equity trade-offs, and guides modal diversification for flood resilience.

Limitations. The absence of a publicly available real-world relief dataset for Central Vietnam required the use of a synthetic instance calibrated from administrative and geographic data. Field validation against actual disaster records remains necessary before operational deployment.

Future work will prioritize three directions. First, constructing a real relief logistics dataset for Central Vietnam in collaboration with local authorities, to enable rigorous empirical validation. Second, rapid practical deployment by integrating the model with government relief planning systems and validating against historical flood response data. Third, strengthening PB-NSGA through improved decoder strategies, adaptive operator tuning, and hybridization with local search to enhance solution quality on large-scale instances.

References

1. Center for Disaster Philanthropy, "2025 southeast asia severe storms," December 2025.
2. Voice of Vietnam, "(translated) natural disasters in 2025: Fierce, abnormal, causing heavy losses in humans and property," December 2025. VOV.VN.
3. N. Altay and W. G. Green III, "Or/ms research in disaster operations management," *European journal of operational research*, vol. 175, no. 1, pp. 475–493, 2006.
4. L. N. Van Wassenhove, "Humanitarian aid logistics: supply chain management in high gear," *Journal of the Operational research Society*, vol. 57, no. 5, pp. 475–489, 2006.
5. Z. Dönmez, B. Y. Kara, Ö. Karsu, and F. Saldanha-da Gama, "Humanitarian facility location under uncertainty: Critical review and future prospects," *Omega*, vol. 102, p. 102393, 2021.
6. C. Boonmee, M. Arimura, and T. Asada, "Facility location optimization model for emergency humanitarian logistics," *International journal of disaster risk reduction*, vol. 24, pp. 485–498, 2017.
7. J. Holguín-Veras, N. Pérez, M. Jaller, L. N. Van Wassenhove, and F. Aros-Vera, "On the appropriate objective function for post-disaster humanitarian logistics models," *Journal of Operations Management*, vol. 31, no. 5, pp. 262–280, 2013.
8. J. F. Campbell, "Integer programming formulations of discrete hub location problems," *European Journal of Operational Research*, vol. 72, no. 2, pp. 387–405, 1994.
9. S. A. Alumur, B. Y. Kara, and O. E. Karasan, "The design of single allocation incomplete hub networks," *Transportation Research Part B: Methodological*, vol. 43, no. 10, pp. 936–951, 2009.
10. O. Sangsawang and S. Chanta, "Capacitated single-allocation hub location model for a flood relief distribution network," *Computational Intelligence*, vol. 37, no. 1, pp. 345–374, 2020.

11. C. Li, P. Han, M. Zhou, and M. Gu, “Design of multimodal hub-and-spoke transportation network for emergency relief under covid-19 pandemic: A meta-heuristic approach,” *Applied Soft Computing*, vol. 133, p. 109925, 2023.
12. S. Tofighi, S. A. Torabi, and S. A. Mansouri, “Humanitarian logistics network design under mixed uncertainty,” *European journal of operational research*, vol. 250, no. 1, pp. 239–250, 2016.
13. M. Yahyaei and A. Bozorgi-Amiri, “Robust reliable humanitarian relief network design: an integration of shelter and supply facility location,” *Annals of Operations Research*, vol. 283, pp. 897–916, 2019.
14. F. S. Salman and E. Yücel, “Emergency facility location under random network damage: Insights from the istanbul case,” *Computers & Operations Research*, vol. 62, pp. 266–281, 2015.
15. K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, “A fast and elitist multiobjective genetic algorithm: Nsga-ii,” *IEEE transactions on evolutionary computation*, vol. 6, no. 2, pp. 182–197, 2002.
16. Y. Zhou, J. Liu, Y. Zhang, and X. Gan, “A multi-objective evolutionary algorithm for multi-period dynamic emergency resource scheduling problems,” *Transportation Research Part E: Logistics and Transportation Review*, vol. 99, pp. 77–95, 2017.
17. M. Gen, F. Altiparmak, and L. Lin, “A genetic algorithm for two-stage transportation problem using priority-based encoding,” *OR Spectrum*, vol. 28, no. 3, pp. 337–354, 2006.
18. C. F. Daganzo, *Logistics systems analysis.* Springer, 2005.
19. P. B. Mirchandani and R. L. Francis, *Discrete location theory.* 1990.
20. E. Zitzler and L. Thiele, “Multiobjective evolutionary algorithms: a comparative case study and the strength Pareto approach,” *IEEE Transactions on Evolutionary Computation*, vol. 3, no. 4, pp. 257–271, 1999.
21. H. Ishibuchi, H. Masuda, Y. Tanigaki, and Y. Nojima, “Modified distance calculation in generational distance and inverted generational distance,” in *Evolutionary Multi-Criterion Optimization*, vol. 9019 of *Lecture Notes in Computer Science*, pp. 110–125, Springer, 2015.
22. F. Barzinpour and V. Esmaeili, “A multi-objective relief chain location distribution model for urban disaster management,” *International Journal of Advanced Manufacturing Technology*, vol. 70, no. 5–8, pp. 1291–1302, 2014.
23. L. Perron and V. Furnon, “Or-tools,” 2025.